

Exercise Sheet 2: Causality

Exercise 3-1: Granger causal test

Year	Chicken (Y)	Egg (X)
1930	468491	3581
1931	449743	3532
1932	436815	3327
1933	444523	3255
1934	433937	3156
1935	389958	3081

a) Auto-regression for X and Y

$$\begin{pmatrix} Y(2) \\ \vdots \\ Y(n) \end{pmatrix} = \beta_0 \begin{pmatrix} 1 \\ \vdots \\ 1 \end{pmatrix} + \beta_1 \begin{pmatrix} Y(1) \\ \vdots \\ Y(n-1) \end{pmatrix} + \beta_2 \begin{pmatrix} X(1) \\ \vdots \\ X(n-1) \end{pmatrix}$$

lag = 1:

$$Y(n) = \begin{pmatrix} 449743 \\ 436815 \\ 444523 \\ 433937 \\ 389958 \end{pmatrix} \quad X(n) = \begin{pmatrix} 3532 \\ 3327 \\ 3255 \\ 3156 \\ 3081 \end{pmatrix}$$

$$Y(n-1) = \begin{pmatrix} 468491 \\ 449743 \\ 436815 \\ 444523 \\ 433937 \end{pmatrix} \quad X(n-1) = \begin{pmatrix} 3581 \\ 3532 \\ 3327 \\ 3255 \\ 3156 \end{pmatrix}$$

Design matrix $\Rightarrow X = \begin{pmatrix} 1 & 468491 & 3581 \\ 1 & 449743 & 3532 \\ 1 & 436815 & 3327 \\ 1 & 444523 & 3255 \\ 1 & 433937 & 3156 \end{pmatrix}$

$$\hat{\beta} = (X^T X)^{-1} X^T Y$$

$$\begin{aligned} (X^T X) &= \begin{pmatrix} 1 & 1 & 1 & 1 & 1 \\ 468491 & 449743 & 436815 & 444523 & 433937 \\ 3581 & 3532 & 3327 & 3255 & 3156 \end{pmatrix} \begin{pmatrix} 1 & 468491 & 3581 \\ 1 & 449743 & 3532 \\ 1 & 436815 & 3327 \\ 1 & 444523 & 3255 \\ 1 & 433937 & 3156 \end{pmatrix} \\ &= \begin{pmatrix} 5 & 2233509 & 16851 \\ 2233509 & 998461944853 & 7535869589 \\ 16851 & 7535869589 & 56922875 \end{pmatrix} \end{aligned}$$

$$(X^T X)^{-1} = A^{-1} = \frac{1}{\det(A)} \text{adj}(A) = \begin{pmatrix} 347,836338 & -1,141 * 10^{-3} & 0,048 \\ -1,141 * 10^{-3} & 4,9775 * 10^{-9} & -3,2132 * 10^{-7} \\ 0,048 & -3,2132 * 10^{-7} & 2,8339 * 10^{-5} \end{pmatrix}$$

$$\hat{\beta} = A^{-1} X^T Y(n) = \begin{pmatrix} 120111,8847 \\ -0,06904307397 \\ 101,3960539 \end{pmatrix}$$

$$RSS_1 = \sum ||e||_2^2$$

$$\begin{aligned} \hat{Y}(n) &= \beta_0 + \beta_1 Y(n-1) + \beta_2 X(n-1) \\ e_n &= Y(n) - \hat{Y}(n) \end{aligned}$$

$$n=2: \hat{Y}(2) = 120111,8847 - 0,069 * 468491 + 101,3961 * 3581 = 450865,0949$$

$$\Rightarrow e_2 = 449743 - 515557,2125 = -1122,094949$$

⋮

$$\Rightarrow e = \begin{pmatrix} -1122,094949 \\ -10376,10786 \\ 17225,49433 \\ 14472,19423 \\ -20199,48642 \end{pmatrix}$$

$$\Rightarrow RSS_1 = 1023104024 = 1,0231 * 10^9$$

b) **Auto-regression without Y**

$$\text{Design matrix without egg feature: } X_0 = \begin{pmatrix} 1 & 468491 \\ 1 & 449743 \\ 1 & 436815 \\ 1 & 444523 \\ 1 & 433937 \end{pmatrix}$$

$$\hat{\beta}_0 = (X_0^T X_0)^{-1} X_0^T Y = \begin{pmatrix} -51719,04695 \\ 1,08061854 \end{pmatrix}$$

$$\hat{Y}_0(n) = \beta_0 + \beta_1 Y(n-1) = -51719,04695 + 1,08061854 * Y(n-1)$$

$$\Rightarrow e = \begin{pmatrix} -4798,013473 \\ 2533,422915 \\ 24211,6594 \\ 5296,251694 \\ -27243,32044 \end{pmatrix} \Rightarrow RSS_2 = 1385892406 = 1,3859 * 10^9$$

c) **Apply statistical test**

$$GS = \frac{\frac{RSS_2 - RSS_1}{d}}{\frac{RSS_2}{n - 2d}} = \frac{1,3859 * 10^9 - 1,0231 * 10^9}{1,3859 * \frac{10^9}{(5 - 2)}} = 0,7853172016 \approx 0,7853$$

$F_{.05}(1, 3) = 10.13$ ([F Distribution Table](#)) $\Rightarrow GS = 0,7853 < 10,13 \Rightarrow$ **Eggs do not G-cause chickens**

d) **Casual test for direction from chicken to eggs**

$$X(n) = \beta_0 + \beta_1 X(n - 1) + \beta_2 Y(n - 1)$$

Design matrix Y: $Y = \begin{pmatrix} 1 & 3581 & 468491 \\ 1 & 3532 & 449743 \\ 1 & 3327 & 436815 \\ 1 & 3255 & 444523 \\ 1 & 3156 & 433937 \end{pmatrix}$

$$\hat{\beta} = (Y^T Y)^{-1} Y^T X$$

$$(Y^T Y)^{-1} = \begin{pmatrix} 347,84 & 0,048 & -1,141 * 10^{-3} \\ 0,048 & 2,834 * 10^{-5} & -3,2132 * 10^{-7} \\ -1,141 * 10^{-3} & -3,2132 * 10^{-7} & 4,978 * 10^{-9} \end{pmatrix} \Rightarrow \hat{\beta} = \begin{pmatrix} -941,4232578 \\ 0,5716433645 \\ 5,11542777 * 10^{-3} \end{pmatrix}$$

$$\Rightarrow e = \begin{pmatrix} 29,83650005 \\ -51,24893533 \\ 60,07020455 \\ -37,20119043 \\ -1,456579013 \end{pmatrix} \Rightarrow \mathbf{RSS_1 = 8511,149774}$$

Design matrix without chicken feature: $\hat{X} = \beta_0 + \beta_1 X(n - 1)$

$$\hat{\beta}_0 = (Y^T Y)^{-1} Y^T X = \begin{pmatrix} 230,7350609 \\ 0,9018648564 \end{pmatrix} \Rightarrow e = \begin{pmatrix} 71,68688833 \\ -89,1217337 \\ 23,76056186 \\ -10,30516848 \\ 3,979452302 \end{pmatrix} \Rightarrow \mathbf{RSS_2 = 13768,29021}$$

$$GS = \frac{\frac{RSS_2 - RSS_1}{d}}{\frac{RSS_2}{n - 2d}} = 1,145488733 \approx 1,145$$

$F_{.05}(1, 3) = 10.13$ ([F Distribution Table](#)) $\Rightarrow GS = 1,145 < 10,13 \Rightarrow$ **Chickens do not G-cause eggs**

$\Rightarrow$ Since neither chickens nor eggs G-cause each other, it's not possible here to say whether what was first, chicken or egg. If we had more data, most probably we could answer that question, but with only 5 usable rows, it wasn't possible.

Exercise 3-2: Multivariate Granger model

- a) Consider graphical Granger model and causal inference by ordinary least squares with adaptive lasso penalty and regularization parameter $\lambda_n = n^{\frac{3}{2}}$. Is the problem solved by this model consistent? Explain your answer in detail.

No, it's inconsistent. Adaptive lasso is consistent under conditions on λ_n :

$$\frac{\lambda_n}{\sqrt{n}} \rightarrow 0 \quad \text{and} \quad \lambda_n n^{\frac{\omega-1}{2}} \rightarrow \infty$$

E.g. for $\lambda_n = \lambda$ constant and $\omega > 3$ fulfilled.

So, we can plug in $\lambda_n = n^{\frac{3}{2}}$ into two limits and evaluate:

$$\frac{\lambda_n}{\sqrt{n}} = \frac{n^{\frac{3}{2}}}{\sqrt{n}} = \frac{n^{\frac{3}{2}}}{n^{\frac{1}{2}}} = n^{\frac{3-1}{2}} = n \rightarrow \infty$$

$$\lambda_n n^{\frac{\omega-1}{2}} = n^{\frac{3}{2}} n^{\frac{\omega-1}{2}} = n^{\frac{3+\omega-1}{2}} = n^{\frac{\omega+2}{2}} \rightarrow \infty \quad (\text{for any } \omega > 0)$$

So, even if the second condition goes to ∞ , the first condition is violated, and for consistency, both conditions should be satisfied.

- b) What is the algorithm HMMLGA for, and what are its hyperparameters?

The HMMLGA algorithm is a genetic algorithm-based search method that explores possible parent sets Q_i for each time series and evaluates them using the HMML (Minimum Message Length) objective. It combines maximum-likelihood estimation with the MML criterion to evaluate candidate causal graphs and iteratively improves them through crossover and mutation ([Hlaváčková-Schindler K, Plant C. Heterogeneous Graphical Granger Causality by Minimum Message Length. Entropy \(Basel\). 2020 Dec 11;22\(12\):1400. doi: 10.3390/e22121400. PMID: 33322439; PMCID: PMC7763266.](#))

Purpose: The HMMLGA algorithm combines hybrid models with Lasso regression for causal inference.

Hyperparameters:

- Population size (m)
- Maximum number of generations (n_g)
- Mutation probability
- Lag parameter (d)
- Ridge regularization parameter (λ)

Exercise 3-3: Bivariate causal models on non-temporal data

- a) Why is the causal relationship between X and Y by bivariate additive noise models for the linear Gaussian case non-identifiable?

Linear-Gaussian models are symmetric, meaning they can fit causal relationships in either direction equally well.

Meaning:

$$Y = aX + N, \quad N \perp X$$

$$X = bY + N', \quad N' \perp Y$$

So, both directions admit an additive noise model with independent Gaussian noise.

Without additional constraints or assumptions, such as non-linearities or external interventions, causal directions cannot be uniquely identified.

- b) Recall the example 1 from lecture 4.

1. What is the causal relation between X and Y?

age (cause) $\rightarrow$ diastolic blood pressure (effect) $\Rightarrow X \rightarrow Y$

This makes perfect sense: as we get older, blood vessels naturally thicken and stiffen over time, increasing the risk of high blood pressure.

2. How one could change $P(\text{effect}|\text{cause})$ without affecting $P(\text{cause})$ in the correct causal direction in this concrete case?

$$P(\text{effect}|\text{cause}) = P(Y|X)$$

$$P(\text{cause}) = P(X)$$

In this concrete situation, we surely can say that age $P(X)$ is the cause. To be able to change $P(\text{effect}|\text{cause})$ without affecting $P(\text{cause})$, we could make people do more sport and have a special diet, or, for example, prescribe them blood pressure medicines. This lowers blood pressure for every age and doesn't change $P(\text{cause})$.

3. Describe how it would be in the anti-causal direction.

Anti-causal direction $\Rightarrow Y \rightarrow X$ (blood pressure causes age)

In that case, we would assume blood pressure causes age. For that, we would have to be able to change how old people are while keeping the overall distribution of blood pressure the same or changing the distribution of blood pressure without changing how ages look inside each blood pressure group. Both ideas don't make any sense. Blood pressure doesn't determine how old the person is, and changing blood pressure doesn't rearrange people's ages – so the anti-causal direction is not possible for this specific case.